

Supervision & Reflection Template

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What?

Reflect on what has happened in your project this month?

This month, I finished one of the most complex functionalities of the project: Smart Insights. What's even more surprising is that this was completed after a previous determination that an intelligent insights-style functionality was not feasible for the project. Nevertheless, after an enlightening discussion with my supervisor, I reevaluated the strategy and concluded that it could be done in a more organized and academically valid manner without having to resort to overly complex AI algorithms.

Smart Insights is now a separate top-level private page and no longer a functionality within Watch Your Watts. The functionality follows a progressive generation model where users click "Generate Insights" and are presented with ranked, high-value recommendations in batches. If additional meaningful insights are available, they can click "Generate More," and once there are no more meaningful recommendations left, the system presents a clear completion message.

On the backend, I built out a run-based insights API to facilitate this progressive experience with a runld system. The insights engine was reworked to be a ranked heuristics engine where it calculates daily kWh and cost from appliance usage data, then generates and scores recommendations based on impact, feasibility, and confidence. I also added in mechanisms such as deduplication, materiality thresholds, and repetitive suggestion limits to avoid presenting users with unrealistic or low-value recommendations. JWT authentication and user-scope access were also maintained for security and isolation purposes.

I also finished the Dashboard component so users can view a summary of their consumption and cost directly after logging in. I also worked on the public landing page to ensure that the entire frontend is more polished and professional.

So What?

Consider what that meant for your project progress. What were your successes? What challenges remain?

The reintroduction of Smart Insights was an important milestone in the project. Initially, I had decided to exclude AI-driven predictions based on complexity and feasibility. However, after a discussion with my supervisor, I understood that it was not the idea of insights that was problematic but the method of implementation. By replacing the AI system with a heuristic engine, I was able to provide smart behavior that is transparent, traceable, and amenable to academic scrutiny.

The current month has further enhanced the analytical capabilities of the project. The system is no longer limited to monitoring data but actually interprets it to offer useful suggestions. The inclusion of the Dashboard also makes the application seem more integrated, as the user now has a convenient entry point to their energy status.

In terms of technical complexity, this has been one of the more challenging integration milestones. The handling of the progressive insights stream, the secure JWT-based user scoping, and the preservation of the frontend and backend contracts were all demanding.

Now What?

What can you do to address outstanding challenges?

The next phase will focus on **testing, debugging, and Sprint 7: Integration & Security Hardening**. This includes:

- Writing and executing structured unit and integration tests.
- Testing edge cases across authentication, insights generation, and Bills Vault access.
- Reviewing security measures such as JWT validation, PIN-protected document access, and input validation.
- Ensuring backend modules are cleanly structured for future PostgreSQL migration.
- Refining UI polish and performance where needed.

At this point, all major features of the system are implemented. The priority now shifts from adding functionality to ensuring robustness, security, and academic-level refinement before final submission.

Student Signature

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